Data Engineer Job Description: Top Duties and Qualifications

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A Data Engineer or Data Systems Engineer, is responsible for developing and maintaining data processing software like databases. Their duties include coordinating with company Executives and other professionals to create unique data infrastructure, running tests on their designs to isolate errors and updating systems to accommodate changes in company needs.

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Data Engineer

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$136,583 / year

Average $136,583

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Data Engineer Duties and Responsibilities

In addition to creating and maintaining an optimal pipeline architecture, typical duties and responsibilities for a Data Engineer position may include:

- Assembling large, complex sets of data that meet non-functional and functional business requirements
- Identifying, designing and implementing internal process improvements including re-designing infrastructure for greater scalability, optimizing data delivery, and automating manual processes
- Building required infrastructure for optimal extraction, transformation and loading of data from various data sources using AWS and SQL technologies
- Building analytical tools to utilize the data pipeline, providing actionable insight into key business performance metrics including

operational efficiency and customer acquisition
- Working with stakeholders including data, design, product and executive teams and assisting them with data-related technical issues
- Working with stakeholders including the Executive, Product, Data and Design teams to support their data infrastructure needs while assisting with data-related technical issues

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Data Engineer Job Description Examples

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What Does a Data Engineer Do?

Data Engineers typically work for corporations in various industries to help them manage and store important data regarding company operations. They work closely with other data professionals and members of the information technology department to design and implement these systems across departments. Their job is to use their knowledge of different programming languages to code and update data systems. They may also be responsible for implementing cybersecurity

Data Engineer Skills and Qualifications

A job description for a Data Engineer should contain the following skills and qualifications requirements:

Copy this section

- Ability to build and optimize data sets, 'big data' data pipelines and architectures
- Ability to perform root cause analysis on external and internal processes and data to identify opportunities for improvement and answer questions
- Excellent analytic skills associated with working on unstructured datasets
- Ability to build processes that support data transformation, workload management, data structures, dependency and metadata

Data Engineer Salary Expectations

- Common salary: **$136,583 yearly**
- Typical salaries range from **$41,000 - $288,000 yearly**
- Find more information on Indeed Salaries

*Indeed data – September 2026

Data Engineer Education and Training Requirements

A Data Engineer position usually requires a hybrid of educational requirements, starting with a degree or certification in information technology or computer science with additional vendor-specific certification.

Google's Certified Professional-Data-Engineer certification establishes that an individual is familiar with the principles related to data engineering and can function as an associate or professional within the industry.

Considered a gold standard by many in the industry, the IBM Certified Data Engineer – Big Data certification focuses on applications specific to big data instead of general skills.

Specific to Cloudera's solutions, the CCP Data Engineer for Cloudera certification shows the individual has proven experience in ETL analytics and tools.

Secondary certifications, including the Microsoft Certified Solutions Expert (MCSE), include a wide range of topics with specific sub-certifications including MCSE: Data Management and Analytics.

Data Engineer Experience Requirements

Many companies look for candidates with more than five years of industry experience who have a graduate degree in information systems, informatics, statistics, computer science or another quantitative field.

Applicants should also have a demonstrated understanding and experience using software and tools including big data tools like Kafka, Spark and Hadoop; relational NoSQL and SQL databases including Cassndra and Pastgres; workflow management and pipeline tools such as Airflow, Luigi and Azkaban; AWS close services including Redshift, RDS, EMR and EC2; stream-processing systems like Spark-Streaming and Storm; and object function/object-oriented scripting languages including Scala, C++, Java and Python.

Data industry events are excellent sources of information and provide an excellent networking opportunities.

Job Description Samples for Similar Positions

If this job description isn't exactly what you're looking for, consider these samples for similar positions:

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- Data Scientist
- Data Analyst
- Data Architect

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Frequently asked questions about Data Engineers

What is the difference between a Data Analyst and a Data Engineer?

The difference between a Data Analyst and a Data Engineer is their areas of job focus. For example, Data Analysts use data systems like databases to pull data regarding customer service, sales quotas, revenue streams or employee satisfaction. In contrast, Data Engineers use their coding skills to develop and update databases and other types of data infrastructure. Data Analysts work closely in meetings with company Executives, IT personnel and department heads to establish the need for new databases or other data infrastructure to aid in data storage. Data Engineers use downtime in their office to code frameworks for new data systems or code updates to existing systems. They also determine the success of new systems or updates by visiting with individual departments and getting feedback.

What are the daily duties of a Data Engineer?

On a typical day, a Data Engineer starts by reviewing upcoming deadlines and checking their email or phone messages to determine changes in assignment needs. Throughout the day, they participate in meetings with company Executives, IT personnel and department heads to establish the need for new databases or other data infrastructure to aid in data storage. Data Engineers use downtime in their office to code frameworks for new data systems or code updates to existing systems. They also determine the success of new systems or updates by visiting with individual departments and getting feedback.

What qualities make a good Data Engineer?

A good Data Engineer uses their knowledge of programming languages in combination with business needs to design, monitor and update data systems for corporations. They also have excellent interpersonal communication, allowing them to speak with employees from a range of departments to address technical problems or database needs. Further, a good Data Engineer always seeks to enhance their coding skills by taking certification courses and participating in training opportunities. A good Data Engineer

also needs to have an investigative mindset that enables them to troubleshoot issues with data systems and isolate defects in data software.



Hire your next Data Engineer today.

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Job Description Examples

Need help writing a job description for a specific role? Use these job description examples to create your next great job posting. Or if you're ready to hire, post your job on Indeed.

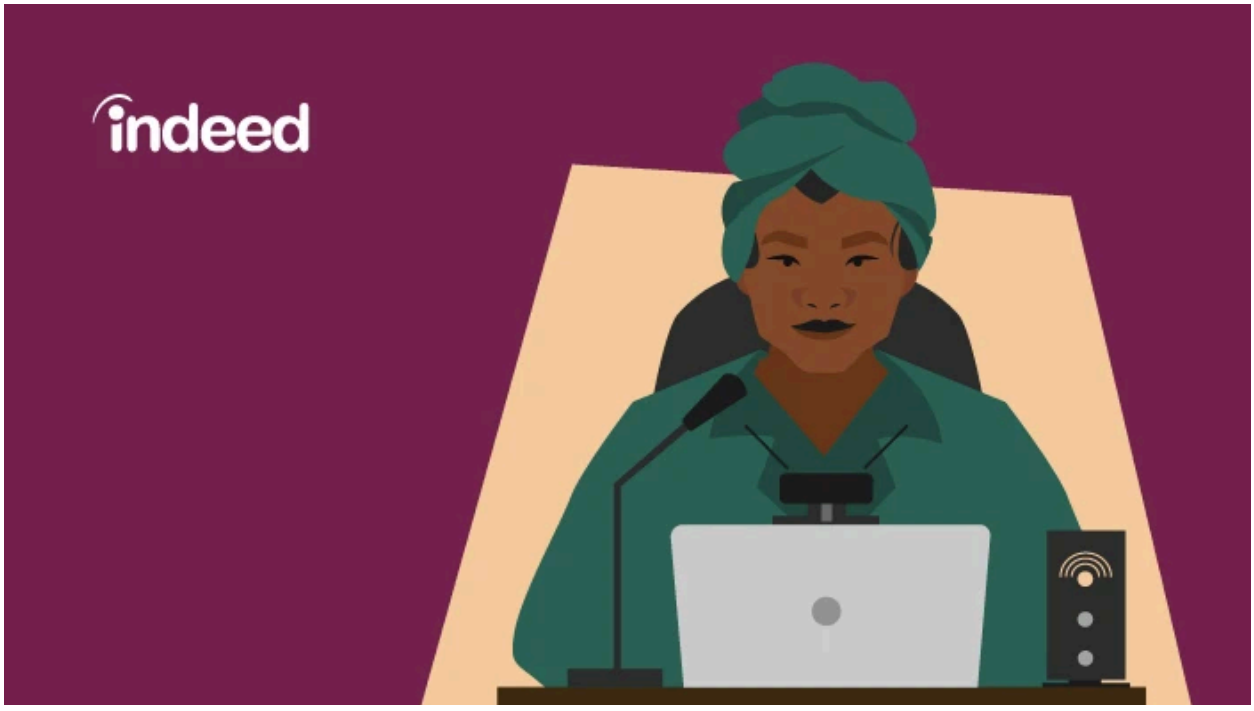
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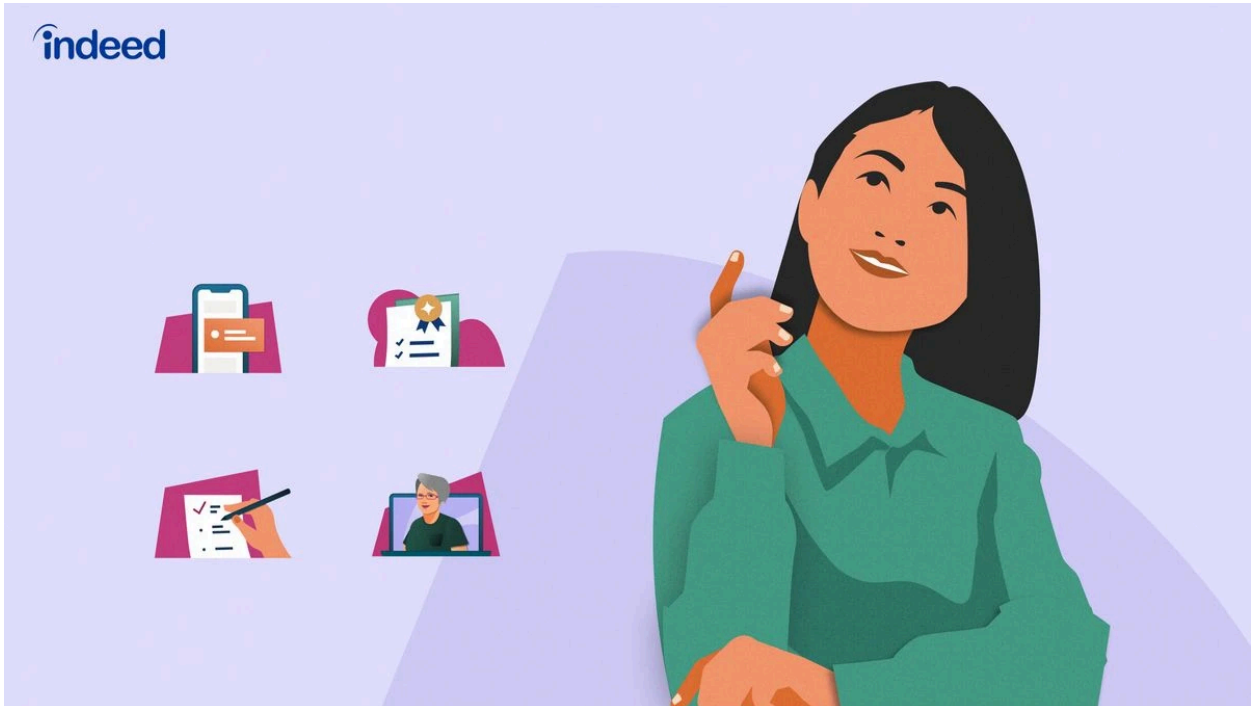
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